

Measuring When a Brain Tumour Segmentation Model Should Not Be Trusted

*A Multi-Institution Generalization, Data-Efficiency and Uncertainty Study on Lower-Grade
Glioma MRI*

Abdul Azeez · AM.SC.U4AIE24049
Saripalli Sri Nandan · AM.SC.U4AIE24043
Dukka Chanakya Tej · AM.SC.U4AIE24070

B.Tech, Computer Science and Engineering (Artificial Intelligence)
Amrita Vishwa Vidyapeetham, Amritapuri Campus

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Abstract

Deep learning models for brain tumour segmentation routinely report high accuracy on a single train/test split, yet give the clinician no indication of when an individual prediction should be doubted. This work treats that omission as the research problem rather than the accuracy figure itself. Using the public TCGA lower-grade glioma cohort (110 patients, 3,929 FLAIR slices, five contributing institutions), we train a U-Net with a ResNet-34 encoder to a tumour-slice Dice of 0.8176, reproducing the 0.82 reported by the dataset authors, and then subject it to three controlled experiments comprising 24 trained models in total.

First, a leave-one-hospital-out protocol paired with volume-matched controls shows that the apparent 0.077 accuracy loss on an unseen institution is not attributable to scanner domain shift: the isolated site effect is -0.0020 with no site reaching statistical significance. Second, a learning curve built from fifteen models across five training-set sizes and three random seeds quantifies the data requirement (fitted ceiling 0.918 Dice, diminishing returns beyond approximately 96 training patients) and establishes a noise floor of 0.057 Dice, below which no effect on this dataset can be resolved. Together these show that per-institution test sets of 14 to 45 patients cannot detect domain shift of the magnitude typically reported, because test-set sampling variance (observed spread 0.131) exceeds the explainable effect (0.032) by a factor of four.

Third, we show that the model can identify its own likely failures. Test-time augmentation over the eight-element dihedral symmetry group yields a disagreement score that correlates with realised Dice at Spearman $\rho = -0.86$; referring the most uncertain 20 per cent of slices to a human raises Dice on the remainder from 0.81 to 0.90. We additionally quantify the limits of this mechanism: approximately 5 per cent of tumour-bearing slices are missed with zero disagreement, a failure mode that disagreement-based uncertainty cannot detect by construction. The work is delivered as a reusable evaluation protocol and a working segmentation system that reports its own confidence.

Keywords: brain tumour segmentation, U-Net, uncertainty quantification, test-time augmentation, domain shift, selective prediction, lower-grade glioma, trustworthy medical AI

1. Introduction

1.1 Motivation

Gliomas are the most common primary malignant brain tumours. Lower-grade gliomas (LGG) appear on fluid-attenuated inversion recovery (FLAIR) MRI as regions of diffuse, irregular hyperintensity whose boundaries are genuinely ambiguous even to trained readers. Manual delineation is slow and shows substantial inter-observer variability, which motivates automated segmentation.

Automated segmentation of this kind is now a solved-looking problem in the literature. Encoder-decoder convolutional networks of the U-Net family reach Dice scores comparable to human readers, and a large number of published and student projects report such figures. The difficulty is that a Dice score is a population statistic. It describes how the model performs on average across a test set; it says nothing about the single scan in front of a clinician. A model that is correct on 82 per cent of cases and confidently wrong on the remainder is indistinguishable, at the point of use, from one that is correct on all of them.

This project therefore takes a different position. We do not attempt to raise the Dice score. We ask three questions about the conditions under which a trained segmentation model can be trusted, and we answer each with a controlled experiment.

1.2 Research questions

- RQ1. Does an unfamiliar hospital degrade segmentation accuracy, and if performance does fall, is the hospital actually the cause?
- RQ2. How much training data does this task require, and how small an effect can a dataset of this size resolve at all?
- RQ3. Can the model identify its own likely failures without access to ground truth, and where does that ability break down?

1.3 Contributions

- A volume-matched control design that separates institutional domain shift from the reduction in training data that necessarily accompanies holding an institution out. To our knowledge this control is absent from prior cross-institutional segmentation studies.
- A measured learning curve and noise floor for the LGG cohort, which together establish the smallest effect this dataset can resolve and show that commonly-sized per-site validation studies are under-powered by roughly a factor of four.
- A validated selective-prediction mechanism built on test-time augmentation, with its operating characteristics and its failure mode both quantified rather than asserted.
- A working demonstration system, TrustSeg, which reports a calibrated confidence verdict alongside every segmentation.

1.4 A note on negative results

Two of the three experiments return negative or corrective findings: the hospital effect we set out to measure is not detectable, and the uncertainty mechanism we validate has a blind spot that covers half of its own failure cases. These are reported as findings rather than as setbacks. A controlled null result on a question the field answers affirmatively by assumption is more informative than another positive result on a saturated benchmark, and the quantified blind spot is what makes the positive uncertainty result honest.

2. Related Work

2.1 Architectures

The U-Net [1] established the encoder-decoder design with skip connections that concatenate encoder feature maps into the decoder, recovering the spatial detail lost to downsampling. Residual learning [2] made deep encoders trainable by allowing gradients to bypass convolutional blocks through identity shortcuts, and ImageNet-pretrained ResNet encoders are now standard in medical segmentation despite the domain gap. nnU-Net [3] demonstrated that systematic configuration of preprocessing and training outperforms most architectural novelty, which informs our decision not to pursue a new architecture.

2.2 Cross-institutional generalization

Zech et al. [4] showed that a pneumonia classifier degraded on external hospital systems and that the source hospital was itself predictable from the image, implying the network had learned site-specific

confounds. AlBadawy et al. [5] examined this directly for brain tumour segmentation using 44 glioblastoma patients from two institutions, finding that same-institution training outperformed cross-institution training. Karani et al. [6] addressed the effect architecturally through domain-specific batch normalisation. Mårtensson et al. [7] documented degradation on out-of-distribution clinical MRI.

All of these studies share a design property that this work targets: removing an institution from training also removes its patients, so site identity and training-set size vary together and the resulting degradation cannot be attributed to either alone.

2.3 Uncertainty quantification

Gal and Ghahramani [8] interpreted dropout at inference as approximate Bayesian inference, and Kendall and Gal [9] separated aleatoric from epistemic uncertainty in vision tasks. The QU-BraTS challenge [10] established scoring criteria for segmentation uncertainty and confirmed that uncertainty estimates carry information beyond the segmentation itself. These methods are evaluated in-domain; their behaviour as detectors of individual-case failure is less often quantified, and their failure modes rarely are.

2.4 The dataset

Buda, Saha and Mazurowski [11] released the LGG segmentation cohort used here and reported a mean Dice of 82 per cent, describing it as comparable to human performance. Their study pooled all five contributing institutions into a single split. The tissue source site is encoded in each patient directory name and is, to our knowledge, unused in subsequent work on this cohort.

3. Dataset and Experimental Setup

3.1 Cohort

We use the LGG Segmentation Dataset [11], obtained from The Cancer Imaging Archive and derived from the TCGA lower-grade glioma collection. It contains 110 patients and 3,929 axial FLAIR slices at 256 x 256 resolution, each with an expert-verified binary mask of the FLAIR abnormality.

Every patient directory is named in the form TCGA_<SITE>_<id>_<date>, where <SITE> is the two-letter TCGA tissue source site code. Parsing this field reveals that the cohort is a latent five-institution multi-site dataset:

Tissue source site	DU	HT	CS	FG	EZ	Total
Patients	45	34	16	14	1	110

Table 1. Patient counts by contributing institution. EZ contributes a single patient and is retained in training in every split; four sites are therefore testable. The 45-to-14 imbalance between DU and FG is the source of the confound addressed in Section 5.

3.2 Model and training configuration

All experiments use an identical configuration; only the split file changes between runs. The network is a U-Net with a ResNet-34 encoder initialised from ImageNet weights, implemented with segmentation-models-pytorch (24,436,369 parameters). FLAIR slices are replicated across three channels to match the pretrained stem and normalised with ImageNet statistics. The network outputs a single-channel logit map; a sigmoid and a threshold of 0.5 produce the binary mask.

Component	Setting	Rationale
Loss	Dice + binary cross-entropy	BCE gives stable per-pixel gradients; Dice optimises overlap directly under severe class imbalance
Optimiser	Adam, initial lr 1e-4	Standard for this architecture
Scheduler	ReduceLROnPlateau, factor 0.5, patience 5	Monitored on validation Dice
Early stopping	Patience 10, max 50 epochs	Prevents overfitting on small splits
Augmentation	Flips, 90-degree rotations, shift-scale-rotate	Geometric only: intensity augmentation would harmonise away the scanner differences under study
Precision	Automatic mixed precision (FP16)	Required to fit 6 GB VRAM
Batch size	4	Hardware constraint

Table 2. Training configuration, held constant across all 24 models.

Training ran on an NVIDIA RTX 4050 laptop GPU with 6 GB of VRAM. This constraint determined the batch size and the use of mixed precision, and ruled out 3D or larger-encoder variants.

3.3 Evaluation protocol

All splits are made at the patient level and asserted leak-free: the split generator fails if any patient appears in more than one partition. Aggregation is per-slice for distribution analysis and per-patient for statistical testing.

Reporting of empty slices. A substantial fraction of slices in this cohort contain no tumour. A model scores a Dice of 1.0 on such a slice by predicting nothing, which inflates the pooled mean by approximately 0.11 on this dataset. We therefore report two figures throughout: Dice over all slices, which is the figure conventionally quoted, and Dice over tumour-bearing slices only, which is the honest measure of segmentation quality. All headline numbers in this report are tumour-bearing Dice unless stated otherwise.

4. Experiment 1: Is the Hospital the Cause?

4.1 Design

The intuitive test for institutional domain shift is leave-one-hospital-out (LOHO): exclude one institution entirely from training and validation, then test exclusively on it. We ran this for the four testable sites.

This design has a confound. Excluding DU removes 45 of 110 patients; excluding FG removes 14. The LOHO models therefore differ from one another in two respects simultaneously - which institution is unseen, and how much training data remains. Any observed degradation is attributable to either.

We therefore added a control arm. For each LOHO split we generated a matched control with identical train, validation and test patient counts, but with test patients sampled randomly across all institutions rather than drawn from one. The control models see the same quantity of data as their LOHO counterparts and differ only in whether an institution is absent. This yields two separable quantities:

$$\begin{aligned}\text{Site effect} &= \text{Dice(LOHO)} - \text{Dice(matched control)} \\ \text{Volume effect} &= \text{Dice(matched control)} - \text{Dice(baseline)}\end{aligned}$$

Nine models were trained for this experiment: one pooled baseline, four LOHO models and four matched controls.

4.2 Results

Experiment	Arm	Test set	Test pts	Slices	Dice (all)	Dice (tumour)	vs baseline
Baseline	A	Mixed	22	760	0.9278	0.8176	—
LOHO — DU	B	DU only	45	1,878	0.8925	0.7052	-0.1125
LOHO — HT	B	HT only	34	1,029	0.8948	0.7420	-0.0757
LOHO — CS	B	CS only	16	358	0.8737	0.7173	-0.1003
LOHO — FG	B	FG only	14	640	0.8733	0.7979	-0.0197
Control — DU	C	Mixed	45	1,605	0.8859	0.6915	-0.1262
Control — HT	C	Mixed	34	1,092	0.8804	0.7268	-0.0908
Control — CS	C	Mixed	16	635	0.9225	0.8228	+0.0052
Control — FG	C	Mixed	14	463	0.8682	0.7291	-0.0885

Table 3. All nine models of Experiment 1. The baseline Dice of 0.8176 reproduces the 0.82 reported by Buda et al. [11] on this cohort, confirming the backbone is correctly trained.

The LOHO column alone appears to support domain shift: accuracy falls on every unseen institution, by a mean of 0.077. The control arm removes that reading. Control — DU, whose test set contains patients from every institution, scores 0.6915 — lower than LOHO — DU at 0.7052. A model that excluded no institution performed worse than one that excluded the largest.

Site	Test pts	Dice LOHO	Dice control	Site effect	Volume effect	p
DU	45	0.7052	0.6915	+0.0137	-0.1262	0.784
HT	34	0.7420	0.7268	+0.0152	-0.0908	0.213
CS	16	0.7173	0.8228	-0.1055	+0.0052	0.337
FG	14	0.7979	0.7291	+0.0688	-0.0885	0.260
Mean	—	0.7406	0.7426	-0.0020	-0.0751	—

Table 4. Decomposition into site and volume effects. *p*-values are Mann-Whitney *U* on per-patient Dice. Note that LOHO and control test sets contain different patients, so this is an independent-samples comparison at matched sample size, not a paired test.

4.3 Interpretation

The mean site effect is -0.0020, which is indistinguishable from zero. Three of the four sites show a positive site effect, meaning the model scored higher on the unseen institution than on a size-matched mixed test set. No site reaches significance at any conventional threshold, the smallest *p*-value being 0.213. The mean volume effect, by contrast, is -0.0751, which accounts for essentially the entire apparent gap.

The conclusion from this experiment alone would be that the degradation is a training-set-size artefact rather than scanner domain shift. Experiment 2 shows that this conclusion is also wrong, and that the true explanation is more interesting.

5. Experiment 2: How Much Data, and How Small an Effect Can Be Resolved?

5.1 Design

To quantify the volume effect directly we constructed a learning curve. The 22 test patients from the baseline split were fixed and reserved; the remaining 88 patients formed a training pool. We sampled five training-set sizes from that pool, and repeated each size with three different random seeds so that the influence of which patients were drawn could be separated from how many. Every one of the fifteen resulting models was evaluated on the same fixed test set, making all numbers directly comparable with the 0.8176 baseline.

5.2 Results

Training patients	Mean Dice	SD across seeds	Fitted	Gain
15	0.6681	0.0474	0.6688	—
30	0.7262	0.0178	0.7223	+0.0581
45	0.7496	0.0273	0.7568	+0.0234
60	0.7862	0.0200	0.7810	+0.0366
75	0.7976	0.0291	0.7989	+0.0114

Table 5. Learning curve, fifteen models. Accuracy climbs steeply to approximately 45 patients and then flattens.

A saturating curve of the form $\text{Dice} = a - b / (n + c)$ fits the five points closely, giving:

$$\text{Dice} = 0.9176 - 13.62 / (n + 39.8)$$

- Projected ceiling as n tends to infinity: 0.918 Dice.
- Diminishing returns beyond approximately 96 training patients, defined as the point at which a further 15 patients yield less than 0.01 Dice. This cohort contains 110 patients in total, so the model is still data-limited at full size.
- Mean standard deviation across seeds: 0.0283. Since these repeats differ only in which patients were drawn, and are evaluated on an identical test set, this is a measurement of training-side noise. Effects smaller than approximately 0.057 Dice (two standard deviations) cannot be resolved on this dataset.

5.3 Placing Experiment 1 on the curve

Each of the nine models from Experiment 1 can now be positioned on the curve by its own training-set size. The residual is the part of its score the curve does not explain.

Model	Arm	Train pts	Observed	Curve predicts	Residual
Baseline	A	72	0.8176	0.7957	+0.0219
LOHO — DU	B	55	0.7052	0.7753	-0.0701
LOHO — HT	B	65	0.7420	0.7875	-0.0455
LOHO — CS	B	80	0.7173	0.8038	-0.0865

Model	Arm	Train pts	Observed	Curve predicts	Residual
LOHO — FG	B	82	0.7979	0.8057	-0.0078
Control — DU	C	55	0.6915	0.7738	-0.0823
Control — HT	C	65	0.7268	0.7875	-0.0607
Control — CS	C	80	0.8228	0.8038	+0.0190
Control — FG	C	82	0.7291	0.8057	-0.0766

Table 6. Residuals against the fitted learning curve.

The mean residual is -0.0525 for the LOHO arm and -0.0502 for the control arm, a difference of -0.0023. This independently reproduces the near-zero site effect of Table 4 by a completely different method.

5.4 The decisive comparison

Quantity	Value
Observed spread across the eight cross-site models	0.1313
Spread the learning curve predicts over that size range	0.0319
Training-side noise floor (two standard deviations)	0.0566

Table 7. The spread between cross-site models exceeds both candidate explanations.

All eight cross-site models trained on between 55 and 82 patients, a region in which the learning curve moves by only 0.032. Their actual scores span 0.131. Neither training-set size nor seed noise can account for a spread four times larger than either.

The remaining variable is test-set composition. The per-site test sets contained between 14 and 45 patients, and which patients they happened to contain swings the measured Dice by more than the entire effect being sought. This yields the principal methodological finding of this work:

Per-institution external validation on fewer than approximately 50 patients cannot detect scanner domain shift of the magnitude reported in the literature, because test-set sampling variance exceeds the effect size by roughly a factor of four. A study of this size that reports a site effect is most likely reporting sampling noise.

6. Experiment 3: Can the Model Flag Its Own Failures?

6.1 Method selection

Monte Carlo Dropout [8] is the standard approach to epistemic uncertainty, but it requires dropout layers active at inference. The segmentation-models-pytorch U-Net contains none, so MC Dropout cannot be applied to these checkpoints without retraining the architecture.

We instead use test-time augmentation (TTA) over the dihedral group D4: the four 90-degree rotations, each with and without a horizontal mirror, giving eight views. Each view is predicted independently and the result mapped back to the original orientation, so all eight predictions should coincide exactly. Their disagreement is the uncertainty signal.

This choice is particularly defensible for this model. Training used horizontal and vertical flips and 90-degree rotations, so the network has every reason to be invariant to precisely these transformations. Disagreement therefore reflects genuine instability in the decision rather than an artefact of presenting an unfamiliar input. We verified by unit test that all eight transforms invert exactly; without this the entire uncertainty analysis would be invalid.

6.2 Candidate scores

Four uncertainty scores were computed on each of the 760 test slices and ranked by Spearman correlation against realised Dice. A strongly negative correlation indicates that higher uncertainty genuinely predicts worse segmentation.

Score	Definition	rho (tumour slices)	rho (all slices)
tta_disagree	1 – mean pairwise Dice between the 8 binarised views	–0.628	–0.885
fg_entropy	Total predictive entropy divided by predicted area	–0.862	+0.606
mean_entropy	Mean pixel-wise entropy of the averaged map	–0.108	–0.778
boundary_ent	Mean entropy over ambiguous pixels ($0.05 < p < 0.95$)	–0.116	–0.883

Table 8. Correlation of each uncertainty score with per-slice Dice ($n = 279$ tumour-bearing, 760 total).

Two scores are retained, for different purposes. Foreground entropy is the stronger ranker where a tumour is present ($\rho = -0.862$), but it divides by predicted area, so on an empty prediction the denominator collapses and the sign inverts. It is therefore valid for offline analysis and unsuitable as a live indicator. View disagreement is weaker on tumour slices but retains its sign on every slice, which is what deployment requires, since one cannot know in advance whether a tumour is present.

6.3 Selective prediction

The operational question is whether referring the most uncertain cases to a human improves accuracy on what remains. Slices were ranked by uncertainty and the most uncertain progressively withheld.

% referred to radiologist	Slices kept	Dice (foreground entropy)	Dice (view disagreement)
0	279	0.8075	0.8075
10	251	0.8817	0.8431
20	223	0.9014	0.8603
30	195	0.9176	0.8673
40	167	0.9262	0.8679
50	140	0.9344	0.8644

Table 9. Retention curves. Referring the most uncertain 20 per cent of slices raises Dice on the remainder from 0.81 to 0.90 under foreground entropy.

Both curves rise, confirming that the model preferentially discards the cases it was about to get wrong. View disagreement plateaus near 40 per cent and then declines, indicating that beyond that point it begins referring cases the model was handling adequately.

6.4 Deployment bands

Thresholds were set at the 60th and 85th percentiles of view disagreement measured on the 279 validated tumour-bearing slices, giving three bands. Validation on 715 slices from 25 patients, selected independently of the test split, confirms monotonic separation.

Verdict band	Slices	Mean Dice
CONFIDENT (disagreement < 0.044)	128	0.938
REVIEW RECOMMENDED (0.044 – 0.141)	46	0.847
HIGH RISK — REFER (≥ 0.141)	28	0.656
HIGH RISK — POSSIBLE MISSED FINDING	11	0.000
NO TUMOUR DETECTED (incorrectly)	11	0.000

Table 10. Band separation on tumour-bearing slices.

6.5 The blind spot

Of 224 tumour-bearing slices examined, 22 (9.8 per cent) are predicted completely empty. These divide cleanly into two groups.

Eleven carry high disagreement: individual views do find the abnormality, but the averaged probability never crosses the 0.5 threshold, so the mask is empty while the views are split. This is the most uncertain state the model can occupy, and the system reports it as a possible missed finding. An earlier version of our demonstration decided the verdict from predicted area before consulting uncertainty, and consequently presented exactly these cases as clean negatives; the ordering was corrected.

The remaining eleven carry disagreement of exactly zero. All eight views unanimously find nothing, and all eight are wrong. There is no signal to detect. This is not an implementation defect but the structural limit of disagreement-based uncertainty: it cannot identify a failure about which the model is uniformly confident. Approximately 5 per cent of tumour-bearing slices fall into this category.

The cost of checking uncertainty before predicted area is a false-alarm rate. On 491 genuinely tumour-free slices, 5.5 per cent are now shown as high risk and a further 0.8 per cent as amber, with 93.7 per cent correctly reported as negative. These false alarms cluster by patient rather than distributing evenly. We regard this as a defensible trade on this task: a false alarm costs a second look, a missed tumour costs more.

7. TrustSeg: The Demonstration System

The experimental findings are delivered as a working application built with Streamlit. A user uploads a FLAIR slice; the system runs the eight-view TTA pass, and returns the segmentation mask, a per-pixel uncertainty heat map, tumour area, hemisphere relative to the estimated brain midline, the number of views that found an abnormality, and a calibrated verdict.

Three design decisions are worth recording, each of which privileges honesty over appearance:

- Uncertainty is evaluated before predicted area. The verdict branches on disagreement first and the mask only selects the wording, so an empty prediction with a split vote is reported as a possible missed finding rather than as a negative.
- Area is reported in pixels by default. The source TIFs carry no pixel spacing metadata, so a figure in square millimetres would be fabricated; spacing can be entered manually if known from the original DICOM.
- Hemisphere is derived from the image's own brain midline and labelled as such. Anatomical lobe-level localisation would require registration to a standard template, which is not performed.

The system was verified through a thirteen-point automated pre-flight check covering dependency versions, checkpoint availability, threshold calibration, exact invertibility of the TTA transforms, the full inference pipeline, and rendering. All thirteen pass.

8. Limitations

- Statistical power. With 14 to 45 test patients per site, Experiment 1 can exclude a large site effect but not a small one. The correct statement is that no effect is detectable at this sample size, not that none exists.
- Attribution. Site identity, scanner vendor and acquisition protocol are inseparable in TCGA metadata. Even a confirmed site effect could not be traced to a specific scanner property.
- The uncertainty blind spot. Approximately 5 per cent of tumour-bearing slices are missed with zero disagreement and cannot be flagged by this mechanism. Addressing them would require an orthogonal uncertainty source, such as an ensemble of independently trained models.
- False alarms. 5.5 per cent of tumour-free slices are flagged as high risk, clustered by patient.
- Scope. Single modality (FLAIR), two-dimensional slice-wise processing, lower-grade glioma only. Volumetric context is not modelled. Tumour type classification is impossible on this cohort, since every patient shares one diagnosis.
- Hardware. 6 GB of VRAM constrained batch size and precision and ruled out larger encoders or 3D architectures.

9. Conclusion

We set out to determine when a brain tumour segmentation model can be trusted, and answered three questions with 24 trained models.

The apparent accuracy loss on an unseen institution is not caused by the institution. A volume-matched control isolates a site effect of -0.0020 with no site reaching significance, and a learning curve reproduces the same near-zero result independently. The true explanation is that per-site test sets of 14 to 45 patients carry sampling variance four times larger than the effect being measured, which means studies of this common size are structurally unable to answer the question they pose.

The data requirement is quantified: a fitted ceiling of 0.918 Dice with diminishing returns beyond roughly 96 training patients, which exceeds the size of this cohort. Any model trained on it remains data-limited.

The model can identify its own failures. Test-time augmentation disagreement correlates with realised Dice at $\rho = -0.86$, and referring the most uncertain fifth of slices raises Dice on the remainder from 0.81

to 0.90. The mechanism's limit is equally well characterised: it cannot detect the approximately 5 per cent of cases about which the model is uniformly and incorrectly confident.

The contribution of this work is not a higher Dice score, and not a new architecture. It is a reproducible protocol for establishing whether a reported generalization result is real, together with a validated mechanism for deciding which individual predictions to trust. Both are applicable to any segmentation model and any dataset.

9.1 Future work

- An ensemble of independently trained models would supply an uncertainty signal orthogonal to test-time augmentation, and would test directly whether the unanimous blind spot is a property of the method or of the cases themselves.
- Conformal prediction would replace the calibrated bands with distribution-free coverage guarantees.
- Repeating the study on a larger multi-site cohort would allow the domain-shift question to be answered rather than shown to be unanswerable at this scale.

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Appendix A: Reproducibility

All experiments are driven by configuration files and are reproducible from the repository. Every split is seed-fixed and asserted leak-free.

Script	Purpose	Models
train.py	Trains one model from a split CSV; --tag isolates outputs	—
make_splits.py	Baseline and four leave-one-hospital-out splits	5
make_control_splits.py	Four volume-matched control splits	4
evaluate_domain_shift.py	Evaluates all nine; site/volume decomposition	—
learning_curve.py	Five sizes $\times$ three seeds on a fixed test set	15
analyze_learning_curve.py	Curve fit, noise floor, residual analysis	—
uncertainty.py	Eight-view TTA, four scores, retention curves	—
app.py	TrustSeg demonstration system	—
smoke_test.py	Thirteen-point pre-flight verification	—

Table A1. Repository contents. Twenty-four models were trained in total.